

- (2 pts) 1. Chloride and sodium are the two most common ions in sea water. List the next four most common ions in sea water, from highest to lowest concentration. Do not list the concentration of each ion. $\frac{1}{2}$ point per ion.

Sulfate, Magnesium, Calcium, Potassium.

- (4 pts) 2. You learned the deepest point in each ocean. List the three *shallowest* points (e.g., trenches) and the ocean in which each point is located. Circle the deepest of the three you list. $\frac{1}{2}$ point per blank plus 1 point for the circle.

Trench Name

Ocean Basin

Eurasian Basin / Litke Deep

Arctic

Sunda/Diamantina

Indian

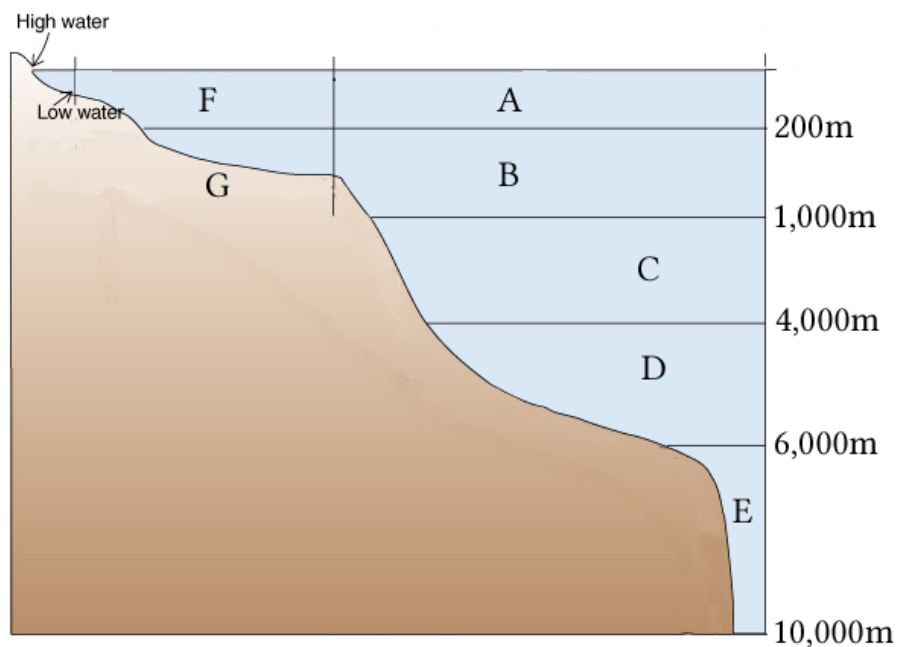
Sandwich

Southern Ocean

- (6 pts) 3. Briefly explain why the planktotrophic larval strategy is less common than the lecithotrophic strategy for marine organisms living at high latitudes.

Planktotrophic larvae eat phytoplankton. The phytoplankton season is so short at high latitudes that a slight mistiming of larvae may miss the food supply. Lecithotrophic larvae do not feed so are not dependent on a food source so they have increased chance of survival.

Use correct names to fill in the blanks below for the zones identified by letters in the figure. (1 point each.)



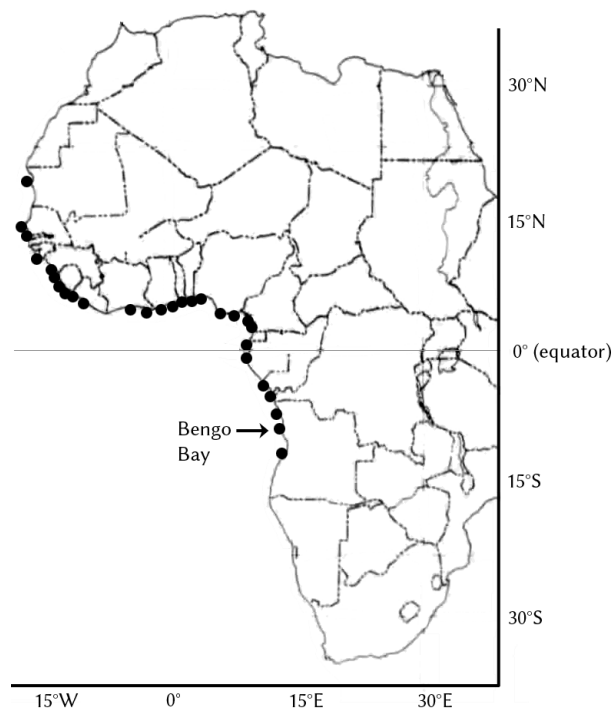
4. **Epipelagic Zone** A, based on depth.
5. **Disphotic Zone** B, based on light.
6. **Bathypelagic Zone** C, based on depth.
7. **Aphotic Zone** C, D, and E combined, based on light.
8. **Neritic Zone** F, open water out to shelf break.
9. **Sublittoral** G, benthic, from low water mark.

Extra credit: Proper names for D and E. (½ point each.)

D. **Abyssalpelagic**

E. **Hadalpelagic**

- (10 pts) 10. The gladiator crab *Callinectes gladiator* lives in estuaries along the west coast of Africa (indicated by the dots). The crab larvae develop over the continental shelf. In spring, the local winds blow from north to south. In fall, the winds blow from south to north. State which season a crab living in Bengo Bay should release its larvae to ensure the larvae reach the shelf and then return to the estuary for metamorphosis to adults. Explain why, using the information provided. Include the role of the Coriolis effect and Ekman transport in your answer.



The site is located in the southern hemisphere.

Therefore, the Coriolis effect causes currents to turn counterclockwise relative to the direction of travel. [2 points for Coriolis and proper direction.]

Ekman transport causes currents to move 90° to the left of the wind direction. [2 points for Ekman and proper direction.]

In fall, winds blow from south to north, so Ekman transport would cause the currents to move out over the continental shelf. This would be the time for the crabs to release its larvae. [3 points]

In spring, the winds move north to south, so Ekman transport would cause the currents to move towards shore. This is when the larvae will return to the estuary. [3 points]

11. You may include illustrations to aid your answers to the following question. Please do not tell me about earthquakes and volcanos in your answers to the following questions.

(5 pts) (a) Name the two types of the Earth's crust, and then list the differences between them for age, density, and thickness.

Continental crust is older, less dense, and thicker than oceanic crust.

(6 pts) (b) Describe ridge push.

1. Magma rises through the space at the edges of two oceanic crust plates, pushing up the plate edges. [2 points]

2. The magma cools on the edge of the plates. [2 points]

3. The weight of the plate edges pushes the plate away from the ridge. [2 points]

(6 pts) (c) Describe slab pull.

1. Denser oceanic crust meets and sinks below the less dense continental crust. [2 points]

2. As oceanic crust sinks, it becomes even denser. [2 points]

3. This causes it to sink faster, pulling along the rest of the plate. [2 points]

- (15 pts) 12. (a) Summarize the classic model of primary productivity in the North Atlantic Ocean from late winter to early summer. Your summary should name the major phytoplankton and zooplankton organisms and describe their roles in overall primary production. Your summary should also describe the seasonal availability of nutrients and light to explain the timing of peak productivity. You may include an illustration. Do not discuss the fall season.

- 1. In the winter, surface temperatures are cold. Winter storms mix surface and deeper water, bringing nutrients to the surface water. [2 points.]**
 - 2. As the winter recedes and spring begins, increasing sunlight triggers the bloom. The diatoms and dinoflagellates undergo a bloom, rapidly increasing primary productivity because they now have both nutrients and sufficient light availability. [2 points. Diatoms must be mentioned.]**
 - 3. During the winter, copepods are down deep. In later spring, the growing adults arrive at the surface, after the bloom and PP has begun. Rapid grazing by copepods (up to 12,000 diatoms per minute) depletes the diatom population, reducing PP. [2 points. Copepods must be mentioned.]**
 - 4. The nutrients at the surface are eventually depleted by remaining diatoms, shutting down PP to very low levels. As fall approaches, the surface cools again. Early winter storms cause mixing, triggering a second smaller bloom. This time, the extent of PP is limited by the shorter days as the fall shifts to winter [2 points]**
- [3 points allotted for clarity, proper terminology, grammar and spelling.]**

- (5 pts) (b) Name and describe the modification to the classic model to account for the amount of primary productivity and retention of POM and DOM in tropical surface waters.

Microbial loop [1 point]

Bacteria consume DOM. Viruses infect and rupture bacteria, releasing OM. [2 points]

Nanozooplankton consume bacteria, transferring OM to higher trophic levels. [2 points]

- (10 pts) 13. Describe the difference between active and passive margins. Describe how this affects the biology of benthic (bottom-dwelling) organisms living on the different margin types. Be clear in your example.